ADP HR Topic Classification - Technical Report

Author: Abraham Solórzano

My Approach & Methodology

The goal was to build a routing engine that accurately directs HR queries to specialized agents. I focused on building a scalable pipeline that ensures data integrity and high reliability for production environments.

Key Assumptions

- **Exclusive Routing:** To keep the user experience simple, I treated this as a multiclass problem where each message gets exactly one destination.
- **Confidence Filtering:** Chatbots shouldn't guess. I implemented a **0.60 confidence threshold**. If the model is unsure, it flags the query as "Unsupported" instead of routing it to the wrong person.
- **Data Integrity & Isolation:** To ensure a fair evaluation, I physically partitioned the dataset into static `train`, `val`, and `test` files before any training occurred. This isolation guarantees that the final metrics are based on completely unseen data, preventing any potential data leakage.

Model Evolution

1. The Baseline (LR + TF-IDF)

I started with a simple Logistic Regression model. It's fast and easy to debug, but it only reached **44% accuracy**. This confirmed that simple keyword matching doesn't work for HR topics because many categories (like Payroll vs Taxes) share the same vocabulary.

2. Deep Learning (Fine-Tuning DistilBERT)

To fix the semantic confusion, I moved to `distilbert-base-uncased`. I chose this because it's much faster than regular BERT while keeping ~97% of its performance. * **Setup:** 10 epochs max, but with `EarlyStopping` to avoid overfitting once the validation loss stops dropping. * **Learning:** Used a 2e-5 learning rate with weight decay for better generalization.

Results & Insights

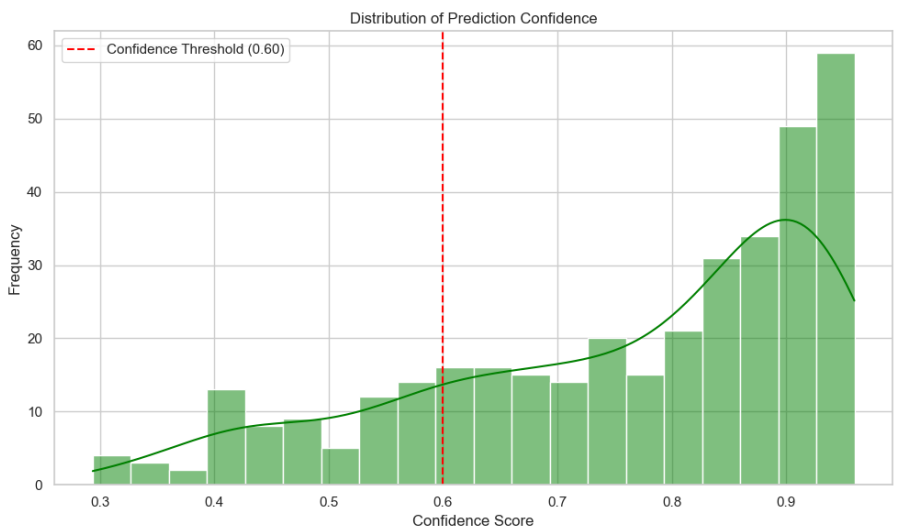
The final model reached **64% accuracy** on the isolated test set. This represents a significant jump from the baseline and provides a solid foundation for a first production release.

Topic Performance breakdown:

- **Tax Services & Benefits (High Accuracy):** The model handles these very well (F1-Scores ~0.70+). The language in these domains is specific enough for the transformer to catch.
- **Payroll (Confusing):** This category shows the most overlap with others. In a future iteration, I'd suggest gathering more niche training samples specifically for payroll.

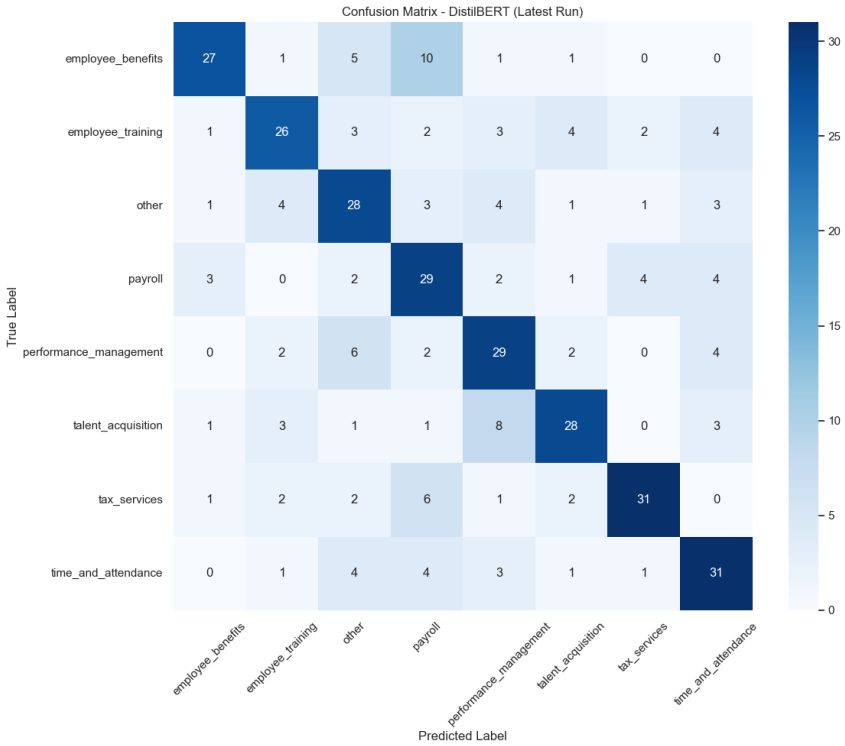
Metrics & Visualization

Confidence Distribution



Most predictions cluster above 0.8, which is great. It means when the model decides, it usually knows what it's doing. The red line shows where we start rejecting low-quality queries to protect the routing accuracy.

Confusion Matrix



You can see the strong diagonal here. The "Other" category is the main source of confusion, which makes sense since it's a catch-all for anything miscellaneous.

Trade-offs & Edge Case Handling

- **Threshold Selection (0.60):** I arrived at this number by analyzing the confidence distribution across the test set. At 0.60, the model effectively filters out-of-domain noise while maintaining a balanced recall for supported topics. In a production setting, this threshold can be tuned dynamically to favor precision over recall (or vice-versa) based on business needs.
- **The "Other" Topic vs "Unsupported":** A key architectural challenge is distinguishing between a supported "Other" query and an "Unsupported" query. Currently, the model uses the softmax probability as a proxy for out-of-domain detection. A future improvement could involve training a dedicated binary classifier for "Supported vs Unsupported" to further improve the safety layer.
- **Class Imbalance:** Given the stratified sampling in our split, the model maintains stable performance. However, for a high-traffic production system, I would recommend implementing class weighting in the loss function to protect underrepresented domains.